

PreflightQC Asset Research: YouTube Video Upload Specifications (August 2026)

TL;DR

- **YouTube publishes almost every encoding parameter — MP4 container, H.264 High Profile, closed GOP, CABAC, 2 B-frames, 4:2:0 chroma, AAC-LC, 48 kHz, the per-resolution bitrate tables — as RECOMMENDED ENCODING, not as hard requirements.** PreflightQC must therefore treat nearly all of these as WARN/INFO and must never FAIL a file for violating them. The only genuine platform-level FAIL/limit conditions are: unsupported container/file type, file size > 256 GB, duration > 12 h (verified) or > 15 min (unverified), and API-enforced metadata limits (title > 100 characters, description > 5000 **bytes**, tags > 500 characters total, empty title).
- **Shorts are classified by YouTube on just two measurable properties: aspect ratio (square or vertical, i.e. $\leq 1:1$) and duration (≤ 3 minutes) for standard-channel uploads on/after 15 October 2024.** YouTube does **not** publish a separate Shorts encoding/codec/bitrate/file-size/minimum-resolution spec, so almost all Shorts-specific technical thresholds must be flagged **UNKNOWN**, not inferred from standard rules.
- **HDR is the only area with concrete, testable technical requirements** (10/12-bit, Rec.2020 primaries + non-constant-luminance matrix, PQ or HLG transfer, and HDR metadata embedded in codec/container). But these are *conditional* rules that apply only when the file is intended as HDR — they are not universal FAIL rules for ordinary SDR uploads.

1. Official Source Register

All pages retrieved and read in full on **9 August 2026** (`hl=en` , US English).

#	Official page title (as published)	URL
1	YouTube recommended upload encoding settings – YouTube Help	https://support.google.com/youtube/answer/1722171?hl=en
2	Supported YouTube file formats – YouTube Help	https://support.google.com/youtube/troubleshooter/2888402?hl=en
3	Video resolution & aspect ratios – Computer – YouTube Help	https://support.google.com/youtube/answer/6375112?hl=en
4	Upload High Dynamic Range (HDR) videos – YouTube Help	https://support.google.com/youtube/answer/7126552?hl=en
5	Upload videos longer than 15 minutes – Computer – YouTube Help	https://support.google.com/youtube/answer/71673?hl=en
6	Understand three-minute YouTube Shorts – YouTube Help	https://support.google.com/youtube/answer/15424877?hl=en
7	Get started creating YouTube Shorts –	https://support.google.com/youtube/answer/10059070?hl=en

	YouTube Help	
8	Add custom thumbnails on YouTube – Computer – YouTube Help	https://support.google.com/youtube/answer/72431?hl=en
9	Supported subtitle and closed caption files – YouTube Help	https://support.google.com/youtube/answer/2734698?hl=en
10	Videos YouTube Data API Google for Developers	https://developers.google.com/youtube/v3/docs/videos
11	Videos: insert YouTube Data API Google for Developers	https://developers.google.com/youtube/v3/docs/videos/insert
12	YouTube Data API - Errors Google for Developers	https://developers.google.com/youtube/v3/docs/errors
13	Choose live encoder settings, bitrates, and resolutions – YouTube Help (LIVE ONLY – NOT upload)	https://support.google.com/youtube/answer/2853702?hl=en
14	Upload 180- or 360-degree videos – YouTube Help	https://support.google.com/youtube/answer/6178631?hl=en
15	Use spatial audio in 360-degree and VR videos – YouTube Help	https://support.google.com/youtube/answer/6395969?hl=en

Non-official sources were consulted for corroboration only and are never treated as authoritative (e.g. Compresto's disputed 512 GB claim; industry guides citing Shorts 1080×1920).

2. Standard YouTube Matrix

Columns: Value → Classification → Source(#) → Confidence → MediaInfo? → ffprobe? → PreflightQC behaviour.

Property	Value / Range	Classification	Src	Conf	MediaInfo	ffprobe	PreflightQC
Container (recommended)	MP4	RECOMMENDED ENCODING	1	HIGH	Yes	Yes	WARN if not MP4
Container / file type (supported set)	.MOV, .MPEG-1, .MPEG-2, .MPEG4, .MP4, .MPG, .AVI, .WMV, .MPEGPS, .FLV, 3GPP, WebM, DNxHR, ProRes, CineForm, HEVC (H.265)	HARD REQUIREMENT (must be on list)	2	HIGH	Yes	Yes	FAIL if not on list
Edit lists	"No Edit Lists (or the video might not get processed correctly)"	RECOMMENDED ENCODING	1	HIGH	Partial	Partial	INFO/WARN

moov atom position	"moov atom at the front of the file (Fast Start)"	RECOMMENDED ENCODING	1	HIGH	Yes	Yes	WARN
Video codec (recommended)	H.264	RECOMMENDED ENCODING	1	HIGH	Yes	Yes	WARN if not H.264
H.264 profile	High Profile	RECOMMENDED ENCODING	1	HIGH	Yes	Yes	INFO/WARN
B-frames	2 consecutive B-frames	RECOMMENDED ENCODING	1	HIGH	Partial	Partial	INFO
GOP	Closed GOP; GOP of half the frame rate	RECOMMENDED ENCODING	1	HIGH	Partial	Partial	INFO
Entropy coding	CABAC	RECOMMENDED ENCODING	1	HIGH	Yes	Yes	INFO
Chroma subsampling	4:2:0	RECOMMENDED ENCODING	1	HIGH	Yes	Yes	INFO
Scan type	Progressive (no interlacing); deinterlace before upload	RECOMMENDED ENCODING	1	HIGH	Yes	Yes	WARN if interlaced
Bitrate mode	Variable bitrate; "No bitrate limit is required"	RECOMMENDED ENCODING	1	HIGH	Yes	Yes	INFO
Video bitrate (SDR, standard FR)	360p 1 / 480p 2.5 / 720p 5 / 1080p 8 / 1440p 16 / 4K 35–45 / 8K 80–160 Mbps	RECOMMENDED ENCODING	1	HIGH	Yes	Yes	INFO (WARN only if grossly low)
Video bitrate (SDR, high FR 48/50/60)	360p 1.5 / 480p 4 / 720p 7.5 / 1080p 12 / 1440p 24 / 4K 53–68 / 8K 120–240 Mbps	RECOMMENDED ENCODING	1	HIGH	Yes	Yes	INFO
Audio codec	AAC-LC or Opus or Eclipsa Audio	RECOMMENDED ENCODING	1	HIGH	Yes	Yes	WARN
Audio channels	Stereo, or Stereo + 5.1, or Stereo + Eclipsa	RECOMMENDED ENCODING	1	HIGH	Yes	Yes	INFO
Audio sample rate	48 kHz	RECOMMENDED ENCODING	1	HIGH	Yes	Yes	INFO
Audio bitrate	Mono 128 / Stereo 384 / 5.1 512 kbps	RECOMMENDED	1	HIGH	Yes	Yes	INFO

	(Eclipsa 128 kbps per channel)	ENCODING					
Standard aspect ratio	16:9 (player auto-adapts to other ratios)	RECOMMENDED ENCODING	1, 3	HIGH	Yes	Yes	INFO
Recommended resolutions (16:9)	240p 426×240; 360p 640×360; 480p 854×480; 720p 1280×720; 1080p 1920×1080; 1440p 2560×1440; 4K 3840×2160; 8K 7680×4320	RECOMMENDED ENCODING	3	HIGH	Yes	Yes	INFO
Max resolution	8K (7680×4320); no playback at intermediate 5K–8K since 2022	UPLOAD LIMIT (playback)	3	HIGH	Yes	Yes	INFO
Minimum resolution	Not stated as a hard rejection floor in official docs	UNKNOWN	—	LOW	Yes	Yes	INFO / DO NOT CHECK
Frame rate	Encode/upload at native; common 24, 25, 30, 48, 50, 60; "other frame rates are also acceptable"	RECOMMENDED ENCODING	1	HIGH	Yes	Yes	INFO
Interlaced source	Deinterlace before upload (e.g. 1080i60 → 1080p30)	RECOMMENDED ENCODING	1	HIGH	Yes	Yes	WARN if interlaced
Padding / black bars	"avoid adding padding or black bars directly to your video"	BEST PRACTICE	3	HIGH	No (not reliably)	No	DO NOT CHECK / INFO
Color space (SDR)	BT.709 (TRC, primaries, matrix)	RECOMMENDED ENCODING	1	HIGH	Yes	Yes	INFO
Pixel aspect ratio	Not explicitly documented; square pixels implied by resolution table	UNKNOWN	—	LOW	Yes	Yes	INFO
	Not documented for SDR (10/12-bit	UNKNOWN					

Bit depth (SDR)	only specified for HDR)	(SDR)	—	LOW	Yes	Yes	INFO
Colour range	YouTube "converts full color range to limited color range"	RECOMMENDED ENCODING (behaviour)	1	HIGH	Yes	Yes	INFO
Max file size	256 GB	UPLOAD LIMIT	5	HIGH	Yes	Yes	FAIL if > 256 GB
Max duration	12 h (verified) / 15 min default (unverified)	UPLOAD LIMIT	5	HIGH	Yes	Yes	WARN / INFO

Verbatim anchors: Source #5 states "The maximum file size you can upload is 256 GB or 12 hours, whichever is less," [Google Support](#) and "By default, you can upload videos that are up to 15 minutes long." [SocialRails](#)

3. Shorts Matrix (researched separately)

Property	Value / Range	Classification	Src	Conf	MedialInfo	ffprobe	PreflightQC
Max duration	3 minutes (standard-channel uploads on/after 15 Oct 2024)	UPLOAD LIMIT (classification)	6	HIGH	Yes	Yes	INFO (drives Shorts classification)
Aspect ratio for classification	"square or vertical aspect ratio" (i.e. $\leq 1:1$)	HARD REQUIREMENT (to be a Short)	6	HIGH	Yes	Yes	INFO
Recommended aspect ratio	9:16 (vertical prioritised)	BEST PRACTICE	7, 15 (ads guidance)	MEDIUM	Yes	Yes	INFO
Recommended resolution	1080×1920 (industry-cited, e.g. BIGVU, Recast Studio)	BEST PRACTICE (non-official)	non-official	LOW	Yes	Yes	INFO
Minimum resolution	Not documented	UNKNOWN	—	LOW	Yes	Yes	DO NOT CHECK
Codec / container / bitrate	No separate Shorts spec published	UNKNOWN	—	LOW	Yes	Yes	DO NOT CHECK
File size limit	Not separately documented (general 256 GB presumed to apply)	UNKNOWN	—	LOW	Yes	Yes	INFO

Music + claim rule	"Shorts longer than one minute that have an active copyright claim, regardless of the policy, will be blocked on YouTube." Google Support	UPLOAD LIMIT (policy)	6	HIGH	No	No	INFO
Official Artist / music-linked channels	Square-or-taller, >1 min uploads: long-form if before 8 Dec 2025; classified as Shorts if on/after 8 Dec 2025	UPLOAD LIMIT (classification nuance)	6	HIGH	Yes (ratio/dur)	Yes	INFO
To avoid Short classification	"use a wider aspect ratio such as 16:9 for long-form videos"	BEST PRACTICE	6	HIGH	Yes	Yes	INFO
#Shorts hashtag	Not required for classification in current docs; classification is by ratio + duration	UNKNOWN / not required	6	MEDIUM	No	No	DO NOT CHECK

Critical Shorts note: Classification is defined by YouTube only in terms of aspect ratio and duration (Source #6: "Any videos uploaded on or after this date with a square or vertical aspect ratio up to three minutes in length will be categorized as Shorts on YouTube."). [Google Support](#) [google](#) Everything else about Shorts encoding is UNKNOWN in official documentation. Do not port standard-upload encoding rules into Shorts FAIL rules.

4. Hard Validation Rules (the ONLY rules that justify a FAIL)

- Unsupported container / file type** — file format not in the 15-item list of Source #2 → YouTube returns "invalid file format." **FAIL**.
- File size > 256 GB** (Source #5) — hard platform ceiling. **FAIL** (or WARN with an explicit "exceeds YouTube maximum" message).
- Title > 100 characters** (Source #10: title "has a maximum length of 100 characters and may contain all valid UTF-8 characters except < and > "). API rejects via `badRequest (400) invalidTitle`. **FAIL (metadata)**.
- Title empty/missing, or snippet update missing title+categoryId** (Source #11/#12: `invalidTitle` ; `invalidVideoMetadata` = "does not set a value for both the snippet.title and snippet.categoryId properties").

GitHub **FAIL (metadata).**

5. **Description > 5000 bytes** (Source #10: description “has a maximum length of 5000 bytes”). google Note: **bytes, not characters.** API error `invalidDescription` . **FAIL (metadata).**
6. **Tags total > 500 characters** (Source #10: tags “has a maximum length of 500 characters”; google commas between items and implied quotation marks around tags containing spaces count toward the limit). API error `invalidTags` . **FAIL (metadata).**
7. **Title or description containing < or >** (Source #10 explicitly excludes these). **FAIL (metadata).**

Everything not on this list should be WARN/INFO/DO NOT CHECK. Note that the numeric limits in #3/#5/#6 live in the **Videos resource property table** (Source #10); the error page only exposes the generic `invalidTitle / invalidDescription / invalidTags` reason codes with no numeric threshold and no `titleTooLong / descriptionTooLong / tagTooLong` codes.

5. Recommendation / Warning Rules (WARN only — never FAIL)

All are RECOMMENDED ENCODING or BEST PRACTICE:

- Container not MP4.
 - Video codec not H.264; profile not High; entropy not CABAC; GOP not closed or $\neq$ half frame rate; not 2 consecutive B-frames; chroma not 4:2:0.
 - Interlaced content present (recommend deinterlace, e.g. 1080i60 $\rightarrow$ 1080p30).
 - moov atom not at front / not fast-start; edit lists present.
 - Audio codec not AAC-LC/Opus/Eclipsa; sample rate $\neq$ 48 kHz; audio bitrate materially below the Mono 128 / Stereo 384 / 5.1 512 kbps recommendations.
 - Video bitrate grossly below the recommended table for the file’s resolution/frame rate (WARN only — YouTube explicitly requires no bitrate limit).
 - SDR colour space not BT.709; RGB matrix used (YouTube discourages it).
 - Added black bars / padding.
 - Duration > 15 minutes (WARN: only uploadable on a phone-verified account).
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6. Informational Properties (report, do not judge)

Exact frame rate; exact resolution/dimensions; measured bitrate (video + audio); GOP length and structure; B-frame count; chroma subsampling; bit depth; colour primaries / matrix / transfer characteristics; colour range (full vs limited); pixel aspect ratio; scan type; audio channel layout; audio sample rate; codec profile/level; container brand/major-brand; duration; file size; presence/position of moov atom; presence of edit lists; number of audio/video tracks.

7. HDR / Colour Handling

SDR (Source #1): YouTube recommends BT.709 as the standard SDR colour space (TRC, primaries, matrix all BT.709). Documented processing behaviour PreflightQC should treat as INFO, not FAIL:

- Standardizes functionally similar matrices/primaries (BT.601 ↔ BT.709) before processing. [google](#)
- Assumes BT.709 TRC when TRC is unspecified; assumes BT.709 matrix/primaries when unknown/unspecified. [google](#)
- Converts unsupported colour spaces to BT.709 by mapping pixel values. [google](#)
- Converts high-bit-depth non-HDR primaries (e.g. BT.2020 without a supported HDR transfer) to BT.709 (8-bit) to avoid banding. [google](#)
- Converts full colour range to limited colour range. [google](#)
- Discourages the RGB colour matrix on uploads. [google](#)

HDR upload requirements (Source #4) — testable, but conditional on HDR intent:

- Resolution: 720p, 1080p, 1440p, or 2160p (prefer UHD widths over DCI, e.g. 3840×1600 not 4096×1716). [google](#)
- Frame rate: 23.976, 24, 25, 29.97, 30, 48, 50, 59.94, 60. [google](#)
- Colour depth: 10-bit or 12-bit. [google](#)
- Colour primaries: Rec.2020.
- Colour matrix: Rec.2020 non-constant luminance. [google](#)
- EOTF: PQ or HLG (Rec.2100).
- Containers tested to work: MOV/QuickTime, MP4, MKV. [google](#)
- Recommended codecs: VP9 Profile 2, AV1, HEVC/H.265. Also work (need very high bitrates): ProRes 422, ProRes 4444, DNxHR HQX, H.264 10-bit. [google](#)
- **HDR metadata MUST be present** in the codec or container or the video will not be processed as HDR. [google](#)
- PQ signalling should carry SMPTE ST 2086 mastering-display metadata + CEA 861-3 MaxCLL and MaxFALL; if missing, YouTube substitutes Sony BVM-X300 values. [google](#)
- Optional dynamic (HDR10+) metadata via ITU-T T.35 terminal codes or SEI headers. [google](#)

PreflightQC HDR behaviour: Evaluate HDR rules only when the file signals HDR intent (PQ or HLG transfer, or Rec.2020 primaries). WARN when an HDR transfer function is present but HDR metadata is missing; WARN when content appears mastered in DCI-P3 but tagged Rec.2020 (Source #4 warns this “will result in an oversaturated look with shifted hues”); [Google Support](#) [google](#) INFO on MaxCLL/MaxFALL presence/absence. **Dolby Vision is not listed among YouTube’s supported HDR upload formats** in Source #4 — flag as UNKNOWN/unsupported rather than validating against it. Live-stream HDR (HEVC over RTMP, HLS) is separate (Source #13) and out of scope for file QA.

8. Conflicts / Unknowns

- **Minimum upload resolution:** No official hard floor. 240p (426×240) is the lowest *recommended encoding* resolution (Source #3) but is not stated as a rejection threshold. → UNKNOWN; do not FAIL.

- **Pixel aspect ratio / anamorphic:** Not officially documented. Square pixels are implied by the resolution table but never mandated. → UNKNOWN.
- **GOP / keyframe interval for uploads:** Documented only as RECOMMENDED (closed GOP, GOP of half the frame rate; Source #1). The oft-quoted “keyframe every 2 seconds” belongs to **live encoder** guidance (Source #13), not uploads. Do not conflate; do not FAIL uploads on GOP length.
- **Shorts encoding specs:** No codec, bitrate, file-size, or minimum-resolution spec is published for Shorts. → UNKNOWN.
- **AV1 (SDR) upload acceptance:** AV1 is explicitly a recommended HDR upload codec (Source #4) and is used across the platform; the supported-formats page (Source #2) lists containers rather than an exhaustive codec set, so general SDR AV1 upload acceptance is inferred at MEDIUM confidence, not stated verbatim.
- **“5000 characters” description myth:** The API states **5000 bytes**, not 5000 characters (Source #10; corroborated by Wordcount, “snippet.description at 5,000 bytes maximum ... non-ASCII characters can use 2–4 bytes each”). [Wordcount](#) Emoji/CJK/Arabic consume multiple bytes, so effective character capacity can be well below 5000. PreflightQC should measure **byte length**, not character count, for descriptions.
- **512 GB file-size claim:** Some third-party guides (e.g. Compresto: “If you verify your account, that jumps to a massive 512GB”) [Compresto](#) assert a 512 GB verified ceiling. This **contradicts** the official page (Source #5: “256 GB or 12 hours, whichever is less”). [Google Support](#) **Trust 256 GB.**
- **Shorts classification date nuance:** Standard channels — on/after 15 Oct 2024. Official Artist Channels / music-Content-Owner-linked channels — square-or-taller, >1 min uploads remain long-form if before 8 Dec 2025, and are classified as Shorts on/after 8 Dec 2025 [Google Support](#) [google](#) (Source #6).
- **Live vs upload separation:** CBR, keyframe interval ≤ 2 s, RTMP/HLS ingestion, and HEVC-over-RTMP for HDR are **live-streaming** settings (Source #13) and must not be applied to file-upload QA.

9. Required / Constrained Metadata Fields

Field	Rule	Classification	Src	PreflightQC
Title (<code>snippet.title</code>)	Required, non-empty; ≤ 100 characters; no < or >	HARD REQUIREMENT	10, 11	FAIL if missing/over/illegal char
Category (<code>snippet.categoryId</code>)	Required together with title on snippet updates	HARD REQUIREMENT	12	FAIL if title set without category (update path)
Description (<code>snippet.description</code>)	Optional; ≤ 5000 bytes ; no < / >	UPLOAD LIMIT	10	FAIL if > 5000 bytes / illegal char
Tags (<code>snippet.tags[]</code>)	Optional; ≤ 500 characters total (commas + implied quotes count)	UPLOAD LIMIT	10	FAIL if > 500 chars total
Default language / audio language	Must be a valid language code	HARD REQUIREMENT	12	INFO (validate against code list if present)
	JPG/GIF/PNG (BMP accepted in some			

Custom thumbnail	help variants); recommended 3840×2160 or 1280×720, min width 640, 16:9; ≤ 2 MB mobile / ≤ 50 MB desktop; Shorts 9:16 at 2160×3840; requires phone-verified channel	UPLOAD LIMIT / BEST PRACTICE	8	FAIL if format unsupported or > size cap; INFO otherwise
Captions / subtitles	SubRip (.srt), SubViewer (.sbv) basic; advanced incl. Scenarist Closed Caption (.scc, preferred for CEA-608), plus other supported markup/broadcast formats	UPLOAD LIMIT (format)	9	FAIL if caption file format unsupported

These metadata rules apply to files delivered via the Data API and to Studio uploads; a desktop QA tool should validate them pre-upload to prevent API/Studio rejections.

10. Machine-Readable Rule Recommendations (schema description — no implementation code)

Each rule record should carry the following fields:

- `id` — stable unique identifier.
- `property` — canonical property path (e.g. `video.codec`, `audio.sample_rate`, `metadata.title.length`).
- `description` — human-readable statement of the rule.
- `classification` — exactly one of: `HARD_REQUIREMENT`, `UPLOAD_LIMIT`, `RECOMMENDED_ENCODING`, `BEST_PRACTICE`, `UNKNOWN`.
- `severity` — exactly one of: `FAIL`, `WARN`, `INFO`, `DO_NOT_CHECK`.
- `scope` — `STANDARD`, `SHORTS`, `HDR`, `METADATA`, or `BOTH`.
- `condition` — operator + expected value/range/enumeration (`equals`, `in_set`, `less_than_or_equal`, `greater_than_or_equal`, `matches`, `present`).
- `applies_when` — guard predicate so conditional rules fire only when relevant (e.g. HDR rules only when transfer ∈ {PQ, HLG} or primaries = Rec.2020; Shorts rules only when aspect ≤ 1:1; description byte-length measured on the raw UTF-8 bytes).
- `detectable_mediainfo` / `detectable_ffprobe` — booleans recording tool coverage.
- `source_url`, `source_title`, `access_date`, `confidence` — provenance for auditability.
- `message_pass` / `message_fail` — display templates.

Governing design principle: severity is *derived from* classification, enforced by a guard so that recommendations can never emit FAIL:

- `HARD_REQUIREMENT` → may map to `FAIL`.
- `UPLOAD_LIMIT` → may map to `FAIL` (or `WARN` when the limit is account-tier-gated, e.g. duration > 15 min).
- `RECOMMENDED_ENCODING` → maps to `WARN` (never FAIL).

- `BEST_PRACTICE` → maps to `WARN` or `INFO`.
 - `UNKNOWN` → maps to `INFO` or `DO_NOT_CHECK` and must surface the “not officially documented” caveat. Add a `source_last_verified` / staleness flag so rules can be re-checked against the live Help pages, since YouTube revises these pages without notice.
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11. Source Pack Status

Well-sourced (HIGH confidence, from official Google/YouTube documentation):

- Supported container/file-type list (Source #2).
- Full recommended upload encoding block — container, codec, profile, GOP/CABAC/B-frames/chroma, audio codec/channels/sample rate, and the complete SDR + HDR bitrate tables and audio bitrate table (Source #1).
- SDR colour-space handling and conversions (Source #1).
- HDR requirements — resolution, frame rate, bit depth, primaries, matrix, EOTF, containers, codecs, and metadata (Source #4).
- Resolution / aspect-ratio recommendations and 8K ceiling (Source #3).
- File size (256 GB) and duration (12 h verified / 15 min unverified) (Source #5).
- Shorts duration (3 min) and aspect-ratio-based classification, plus the >1-min copyright-claim block and the Official-Artist-Channel 8 Dec 2025 nuance (Source #6).
- Metadata field limits — title 100 characters, description 5000 bytes, tags 500 characters, `</>` exclusion, and the `invalidTitle` / `invalidDescription` / `invalidTags` / `invalidVideoMetadata` error reasons (Sources #10, #11, #12).
- Custom thumbnail specs (Source #8) and supported caption/subtitle formats (Source #9).

Partially sourced (MEDIUM confidence):

- General SDR AV1 upload acceptance (explicit only for HDR).
- Shorts recommended 9:16 / 1080×1920 (official for ads/general guidance; the exact resolution figure is industry convention, not a published Shorts spec).

UNKNOWN / not officially documented (flag, do not guess):

- Minimum upload resolution as a hard rejection floor.
- Pixel aspect ratio / anamorphic handling.
- SDR bit-depth requirement.
- Shorts codec/bitrate/file-size/minimum-resolution specs.
- Upload-side keyframe interval as a hard rule.
- Dolby Vision upload support.
- Whether `#Shorts` hashtag affects classification (current docs classify by ratio + duration only).

Explicitly excluded from FAIL logic:

- All live-streaming encoder settings (CBR, keyframe ≤ 2 s, RTMPS/HLS, HEVC-over-RTMP HDR) — documented

in Source #13 but **LIVE-only**, not applicable to local-file upload QA.

Staleness caveat: All values reflect the official pages as read on 9 August 2026. YouTube edits these Help/API pages without changelogs; PreflightQC should re-verify the encoding-settings, file-formats, HDR, Shorts, and Data API pages periodically and treat any rule whose source has since changed as provisional until re-confirmed.